

HalluGen: Synthesizing Realistic and Controllable Hallucinations for Evaluating Image Restoration

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Abstract

Generative models are prone to hallucinations: plausible but incorrect structures absent in the ground truth. This issue is problematic in image restoration for safety-critical domains such as medical imaging, industrial inspection, and remote sensing, where such errors undermine reliability and trust. For example, in low-field MRI, widely used in resource-limited settings, restoration models are essential for enhancing low-quality scans, yet hallucinations can lead to serious diagnostic errors. Progress has been hindered by a circular dependency: evaluating hallucinations requires labeled data, yet such labels are costly and subjective. We introduce HalluGen, a diffusion-based framework that synthesizes realistic hallucinations with controllable type, location, and severity, producing perceptually realistic but semantically incorrect outputs (segmentation IoU drops from 0.86 to 0.36). Using HalluGen, we construct the first large-scale hallucination dataset comprising 4,350 annotated images derived from 1,450 brain MR images for low-field enhancement, enabling systematic evaluation of hallucination detection and mitigation. We demonstrate its utility in two applications: (1) benchmarking image quality metrics and developing Semantic Hallucination Assessment via Feature Evaluation (SHAFE), a feature-based metric with soft-attention pooling that improves hallucination sensitivity over traditional metrics; and (2) training reference-free hallucination detectors that generalize to real restoration failures. Together, HalluGen and its open dataset establish the first scalable foundation for evaluating hallucinations in safety-critical image restoration. The code and dataset are available at <https://github.com/edshkim98/HalluGen>.

1. Introduction

Hallucinations, visually plausible yet incorrect content, are a fundamental failure mode of modern generative models

for image restoration. These errors are critical in safety-critical domains where restoration is a prerequisite for expert analysis (e.g., radiological reporting, human QC) and automated downstream tasks (e.g., biomarker estimation via FreeSurfer). In such human-in-the-loop and algorithmic workflows, hallucinations arising from generative uncertainty can lead to catastrophic misinterpretation.

Despite recent advances in image restoration [8, 26, 31, 32, 37, 45], these models still produce hallucinations that are difficult to detect or measure. As shown in Fig. 1 (top), progress is limited by a *circular dependency*: hallucination analysis requires labeled data, yet hallucinations are ambiguous and costly to annotate. Measuring hallucinations is non-trivial as common metrics (PSNR, SSIM, LPIPS [53]) favor perceptual sharpness over correctness, often scoring hallucinated outputs highly. Existing evaluations rely on downstream task performance [21–23] or manual labels [2, 35], which either conflate hallucinations with task accuracy or are restricted to simple datasets (e.g., MNIST [28], Hands [1]).

To break this deadlock, we introduce *HalluGen*, a diffusion-based framework for controllable and realistic hallucination generation. Following recent work [6, 23, 24], we categorize hallucinations as (1) *intrinsic*, which break measurement consistency, and (2) *extrinsic*, which remain consistent but are semantically incorrect.

HalluGen extends diffusion-based inverse solvers with targeted gradient perturbations, enabling explicit control over hallucination *type, location, and severity*. Controllability is crucial as hallucinations in decision-critical regions (e.g., lesions, solder joints) pose the greatest risk. To the best of our knowledge, no prior work enables controllable, taxonomically grounded hallucination synthesis for systematic hallucination analysis in image restoration. We instantiate HalluGen with Diffusion Posterior Sampling [10], though it is compatible with other diffusion formulations.

Our contributions are threefold:

1. We propose *HalluGen*, a diffusion-based framework for controllable hallucination generation in image restoration, generalizing across medical (HCP) [42], industrial

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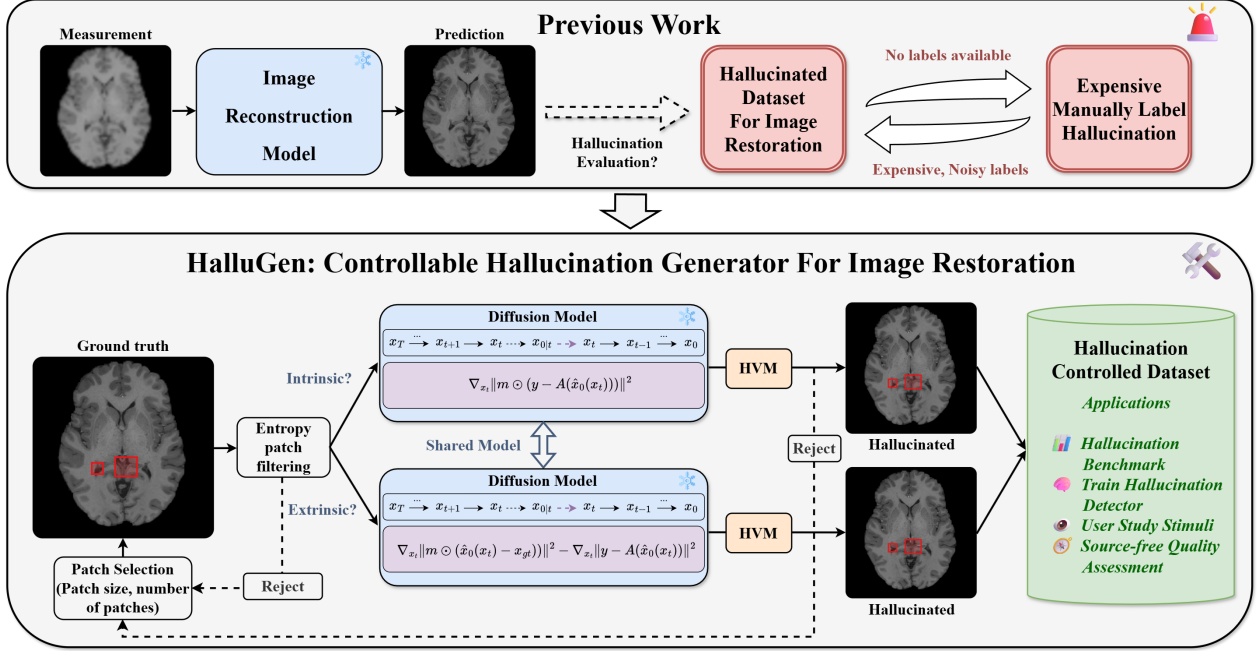


Figure 1. **Circular dependency in hallucination evaluation and our proposed HalluGen solution.** Top: Reliable hallucination analysis requires labeled data, but obtaining labels demands expert annotation with high disagreement. Bottom: HalluGen breaks this loop by generating controllable hallucinations with automatic labels, enabling systematic benchmarking, and perceptual studies across domains.

- (MVTec AD [5]), and natural (ImageNet [12]) images.
- Using HalluGen, we construct the *first large-scale hallucination dataset* for low-field MRI enhancement, comprising 4,350 predictions with patch-level annotations across 1,450 images. This dataset establishes a foundation for systematic hallucination analysis.
- Using the HalluGen-generated dataset, we demonstrate two key applications: (a) a *hallucination metric benchmark*, revealing that existing metrics overlook hallucinations. We develop *Semantic Hallucination Assessment via Feature Evaluation (SHAFE)*, a feature-based metric with soft-attention pooling that improves hallucination detection AUC by 0.25 and reduces false negatives by 24 percentage points; and (b) *reference-free hallucination detectors* trained on HalluGen and evaluated on real restoration outputs without ground-truth references.

Together, HalluGen and our dataset establish the first systematic foundation for generating and evaluating hallucinations in image restoration, advancing toward safer and more reliable deployment in safety-critical applications.

2. Related Work

2.1. Image Restoration

Image restoration aims to recover clean, high-quality images from degraded measurements across tasks such as super-resolution, denoising, and deblurring. Modern supervised models learn powerful data-driven priors for per-

ceptual quality [8, 14, 31, 32, 45], while recent works leverage pre-trained diffusion models as generative priors for inverse problems [10, 11, 20, 39, 46, 51, 55]. These approaches have been extended to medical imaging, including compressed sensing and low-field MRI enhancement [3, 15, 18, 23, 27, 33, 43]. However, emphasizing perceptual realism can induce hallucinations, plausible but incorrect structures, reflecting the perception–distortion trade-off [7, 45]. In safety-critical domains, hallucination is a known risk but remains largely unstudied, with only a few recent work [2, 22, 23, 35] proposing mitigation strategies.

2.2. Image Quality Metrics

Traditional full-reference metrics [40, 48, 54] focus on pixel-wise fidelity and global structural similarity, making them sensitive to misalignment and ineffective at capturing localized or semantically incorrect errors. Learned perceptual metrics [13, 53] improve correlation with human perception by comparing deep feature representations, but they aggregate errors globally and remain insensitive to spatially sparse hallucinations. In medical image reconstruction, recent efforts to quantify hallucinations [6, 44] are restricted to linear forward models or task-specific architectures, limiting generality. These limitations motivate the SHAFE metric (Section 4.5), which computes local (e.g. patch) metric maps and aggregates them using soft-attention weighting, increasing sensitivity to localized hallucinations.

2.3. Hallucination Benchmarks

Recent benchmarks for language and vision-language models (LLMs and VLMs) evaluate semantic hallucinations such as factually incorrect or unverifiable outputs. Examples include HaluEval, HalluLens, and HallusionBench for text–image reasoning [4, 16, 29], POPE for object hallucination [30], and VideoHalluciner for temporal hallucination analysis [47]. While some recent works [19, 36, 49] attempt to synthesize hallucinations to automate annotation in VLMs, they lack spatial control (e.g., size, location) over how hallucinations are generated. To our knowledge, no benchmark exists for hallucinations in image restoration with spatial annotations, a gap our work addresses.

3. Method

3.1. Problem Formulation

Image restoration aims to recover a high-quality image $x \in \mathbb{R}^N$ from its degraded observation $y \in \mathbb{R}^M$, formulated as an inverse problem:

$$y = \mathcal{A}(x) + n, \quad (1)$$

where $\mathcal{A}(\cdot)$ denotes the forward degradation operator and n is measurement noise. As $\mathcal{A}$ is typically non-invertible, recovery is ill-posed.

Hallucination Taxonomy. We categorize hallucinations based on their relationship to the forward operator $\mathcal{A}(\cdot)$, following the recent work on hallucination [6, 23].

Intrinsic hallucinations violate measurement consistency, introducing features inconsistent with observed data:

$$\mathcal{A}(\hat{x}) \neq \mathcal{A}(x_{\text{gt}}), \quad (2)$$

where $\hat{x}$ and x_{gt} denote the reconstructed image and ground truth, respectively.

Extrinsic hallucinations preserve measurement consistency but differ in the inverse domain of $\mathcal{A}$:

$$\mathcal{A}(\hat{x}) = \mathcal{A}(x_{\text{gt}}), \text{ but } \mathcal{A}^{-1}(\mathcal{A}(\hat{x})) \neq \mathcal{A}^{-1}(\mathcal{A}(x_{\text{gt}})). \quad (3)$$

This distinction is crucial: intrinsic hallucinations can be detected through measurement consistency checks, while extrinsic hallucinations require ground truth or domain knowledge. In medical imaging, both pose serious risks as they may obscure clinically significant details. Understanding and systematically evaluating both categories motivates the need for controlled hallucination generation.

3.2. Why do we need hallucination generator?

Existing metrics are insensitive to hallucinations. Figure 2 shows that standard image quality metrics (PSNR, SSIM, LPIPS) assign higher scores to hallucinated predictions from pretrained conditional diffusion models on

MVTec AD [5] and BraTS [34] than to slightly blurred but semantically correct images. This exposes a critical flaw: these metrics favor perceptual sharpness over correctness of content, failing to penalize plausible yet false features. Such bias highlights the need for hallucination-aware evaluation, which requires labeled hallucination data, a challenging circular dependency between generation and evaluation.

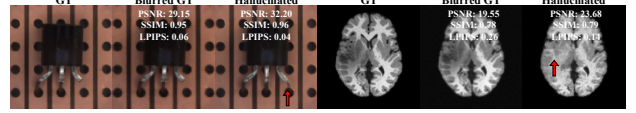


Figure 2. **Existing metrics fail to penalize hallucinations.** Across MVTEC AD (left) and BraTS (right), PSNR, SSIM, and LPIPS assign higher scores to hallucinated predictions than to slightly blurred but correct images, reflecting a bias toward perceptual sharpness over correctness.

Manual labeling is unreliable and uncontrolled. An intuitive approach to hallucination evaluation is manual annotation, but it is neither scalable nor reliable. To verify this, two domain experts annotated hallucination regions at the *patch level* across 50 images, yet the inter-annotator agreement was low (Cohen’s $\kappa = 0.30$), far below the accepted threshold ($\kappa > 0.60$). This highlights the subjective and ambiguous nature of spatial hallucination identification. These limitations motivate *HalluGen*, which synthesizes realistic hallucinations with known ground truth, enabling reproducible and scalable benchmarking.

3.3. HalluGen

Background: Diffusion Posterior Sampling. We build on denoising diffusion probabilistic models (DDPMs) [17] as our generative prior. Given a clean image x_0 , the forward process gradually adds Gaussian noise:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I), \quad (4)$$

where x_t denotes the noisy image at timestep t , $\bar{\alpha}_t = \prod_{i=1}^t(1 - \beta_i)$, β_t is the noise schedule. The reverse process denoises from $t = T$ to 0 using a learned model, where T is the total number of diffusion timesteps.

To solve inverse problems, Diffusion Posterior Sampling (DPS) [10] injects data consistency into the reverse process. At each step, a gradient update enforces the measurement constraint $y = \mathcal{A}(x)$:

$$x_{t-1} = \mu_\theta(x_t, t) - \lambda_t \nabla_{x_t} \|y - \mathcal{A}(\hat{x}_0(x_t))\|^2 + \sigma_t \epsilon, \quad (5)$$

where μ_θ is the model-predicted mean, $\hat{x}_0(x_t)$ is the Tweedie estimate of the clean image, λ_t controls guidance strength, and $\epsilon \sim \mathcal{N}(0, I)$. This steers sampling toward solutions consistent with the measurement y .

Intrinsic Hallucination Generation. HalluGen enables spatially controllable hallucination synthesis by operating

at the patch level. It samples random binary masks $m \in \{0, 1\}^{H \times W}$ that define injection regions, varying patch sizes to create hallucinations at different spatial scales. To induce intrinsic hallucinations that violate measurement consistency, HalluGen applies gradient *ascent* on the data-consistency term within masked regions (pushing away from consistency) and gradient *descent* outside. This process can be formulated as:

$$x_{t-1} = \mu_\theta(x_t, t) - \lambda_t \nabla_{x_t} \|(1 - m) \odot (y - \mathcal{A}(\hat{x}_0(x_t)))\|^2 + \gamma_t \cdot \nabla_{x_t} \|m \odot (y - \mathcal{A}(\hat{x}_0(x_t)))\|^2, \quad (6)$$

where $\gamma > 0$ controls the ascent strength. This formulation creates measurement space violations $\mathcal{A}(\hat{x}) \neq \mathcal{A}(x_{\text{gt}})$ specifically within the masked patches, generating intrinsic hallucinations while preserving fidelity elsewhere.

Extrinsic Hallucination Generation. Extrinsic hallucinations occur when reconstructions deviate in regions imperceptible to the forward operator while preserving measurement consistency. For general (nonlinear) operators $\mathcal{A}(\cdot)$, such regions, analogous to a null space, cannot be computed explicitly. HalluGen therefore applies gradient *ascent* in the ground-truth image space to induce semantic deviations while maintaining measurement fidelity.

However, pixel-space divergence alone yields suboptimal results, as the optimization seeks a compromise between three constraints: the learned data distribution $p_{\text{data}}(x)$, measurement consistency, and ground-truth divergence. To address this issue, we incorporate feature-space divergence using a pretrained feature extractor $F(\cdot)$ (e.g., DINO [41], SAM [25], MedSAM [9]):

$$x_{t-1} = \mu_\theta(x_t, t) - \lambda_t \odot \nabla_{x_t} \|y - \mathcal{A}(\hat{x}_0(x_t))\|^2 + \gamma_{1,t} \cdot \nabla_{x_t} \|m \odot (\hat{x}_0(x_t) - x_{\text{gt}})\|^2 + \gamma_{2,t} \cdot \nabla_{x_t} \|m \odot (F(\hat{x}_0(x_t)) - F(x_{\text{gt}}))\|^2, \quad (7)$$

where $\gamma_{1,t}$ and $\gamma_{2,t}$ control pixel- and feature-space divergence strengths at time step t . This generates semantically meaningful hallucinations that maintain visual realism while deviating substantially in feature space.

Hallucination Verification Module (HVM). To ensure hallucinations remain visually plausible yet perceptually distinct, we measure their effect size using Cohen’s d within hallucinated regions m at the final diffusion step ($t = 0$). Samples not meeting the taxonomy criterion are re-sampled until all patches satisfy the threshold τ_{hvm} .

For **intrinsic** hallucinations, we enforce a measurement-domain violation:

$$d_{\text{meas}} = \frac{\mu_{\text{meas}}^{\text{pred}} - \mu_{\text{meas}}^{\text{gt}}}{\sigma_{\text{meas}}} \geq \tau_{\text{hvm}}. \quad (8)$$

For **extrinsic** hallucinations, we require measurement consistency but image-domain deviation:

$$\begin{cases} d_{\text{meas}} \leq \tau_{\text{hvm}}, \\ d_{\text{img}} = \frac{\mu_{\text{img}}^{\text{pred}} - \mu_{\text{img}}^{\text{gt}}}{\sigma_{\text{img}}} \geq \tau_{\text{hvm}}, \end{cases} \quad (9)$$

where $\mu_{\text{meas}}^{\text{pred}}, \mu_{\text{meas}}^{\text{gt}}, \mu_{\text{img}}^{\text{pred}}$, and $\mu_{\text{img}}^{\text{gt}}$ are masked group means, and σ_{meas} and σ_{img} are the corresponding pooled standard deviations. Cohen’s d provides a normalized effect size, making HVM domain-agnostic. τ_{hvm} controls how strictly generated samples must adhere to the taxonomy definitions.

Manifold Regularization Effect. Although gradient ascent perturbs the diffusion process, generated images remain realistic due to the diffusion prior. The denoising network μ_θ acts as a soft manifold prior, projecting samples toward high-likelihood regions of $p_{\text{data}}(x)$ in each time step. This preserves global coherence while introducing localized deviations, yielding plausible yet incorrect structures. To further enhance realism, we gradually suppress gradient ascent in the final steps, allowing the denoising prior to dominate as the process converges toward the data manifold.

Implementation Details. A key design choice is *where* to inject hallucinations as they typically occur in non-homogeneous regions. Random sampling often hits background, while segmentation-based masking needs annotations. We instead use an entropy-based strategy that selects informative regions, with patch sizes randomly drawn from 16–24 pixels to match typical hallucination extent. For each candidate patch, we compute its Shannon entropy:

$$H(p) = - \sum_i p_i \log p_i, \quad (10)$$

where p_i is the normalized intensity histogram. Patches failing the entropy or background-ratio thresholds (e.g., out-of-brain regions) are rejected and resampled. This lightweight, annotation-free heuristic reliably targets semantically meaningful regions.

3.4. Dataset Construction

A hallucination dataset is constructed using HalluGen on 1,450 MRI slices. A diffusion model is first trained on high-quality 3T brain MR images from the Human Connectome Project (HCP) [42], each with resolution 256×256 . To emulate low-field MRI ($<0.36\text{T}$), following DynamicDPS [23], a composite degradation operator is applied, $\mathcal{A} = \text{Blur}(\text{DS}_k(\Gamma_\gamma(\cdot)))$, with $k = 4$ and $\gamma = 0.7$, creating a moderately ill-posed nonlinear inverse problem suitable for controlled hallucination generation.

For each ground-truth image x_{gt} , three outputs are generated: (1) a non-hallucinated baseline using DPS; (2) intrinsic hallucinations; and (3) extrinsic hallucinations syn-

thesized by HalluGen. Each hallucinated image contains $n \in \{1, 2, 3\}$ non-overlapping patches.

Outside masked regions are smoothly interpolated with the baseline at each diffusion timestep to confine evaluation to injected hallucinations. Each entry consists of $(x_{\text{gt}}, y, \hat{x}, m)$ and the dataset is balanced across types (1,450 intrinsic and extrinsic cases each).

4. Experiments

4.1. Dataset Statistics

The dataset maintains balanced coverage across hallucination characteristics. Intrinsic and extrinsic samples each account for 50% of the dataset, and images with one to three hallucinated patches are evenly distributed (30–38%). Spatially, hallucinations span all four image quadrants (22–30%), ensuring diverse anatomical coverage. This balanced design supports fair benchmarking across diverse scenarios.

4.2. Validation of Generated Hallucinations

Figure 3 shows intrinsic and extrinsic hallucinations generated by HalluGen. The images appear realistic, altering tissue morphology such as sulci and ventricles. A key trend emerges: intrinsic hallucinations often cause morphological distortions (shape or size), whereas extrinsic ones introduce new plausible structures, leading to topological changes.

We assess perceptual realism using FID and semantic deviation via tissue segmentation IoU within hallucinated regions. Since no prior work addresses hallucination generation for image restoration, we compare HalluGen against three baselines: (1) vanilla DPS [10] as a non-hallucinated reference, (2) random rotation in selected patches, and (3) measurement perturbation by adding noise in measurement space. We additionally evaluate HalluGen with different feature extractors. Gaussian Mixture Models are used for segmenting gray matter, white matter, and CSF.

Table 1 shows HalluGen achieves FID comparable to DPS, while segmentation IoU drops by 50% within hallucinated regions, indicating strong semantic deviation. Although random rotations and measurement perturbations yield low FID, they apply only to intrinsic cases and introduce patch artifacts. HalluGen performs consistently across different feature extractors; additional visuals are provided in the supplementary. An expert study ($n=2$), where HalluGen outputs were mixed with vanilla DPS reconstructions, yielded only 50.5% identification accuracy, confirming that HalluGen produces visually realistic hallucinations.

We validate HalluGen’s taxonomy compliance in Tab. 2 by measuring measurement- and image-space deviations within masked regions. Intrinsic hallucinations exhibit measurement loss roughly 7 times higher than DPS, confirming violation of measurement consistency. Extrinsic hallucinations retain low measurement loss but diverge in

Table 1. **Comparisons of perceptual realism and semantic deviation of HalluGen and other baselines.** For HalluGen, FID is low, comparable to DPS, confirming realism, while segmentation IoU within hallucinated regions drops sharply across different feature extractors, verifying semantic deviation.

Method	FID ↓	IoU ↓	Tax. Comp.
DPS [10]	0.32	0.861 ± 0.09	-
Random Rotation	0.32	0.393 ± 0.12	Intr. only
Meas. Perturb.	0.48	0.721 ± 0.10	Intr. only
HalluGen + MedSAM [9]	0.41	0.363 ± 0.15	Both
HalluGen + SAM [25]	0.36	0.367 ± 0.15	Both
HalluGen + DINOv3 [41]	0.43	0.362 ± 0.04	Both

image space, consistent with the formal taxonomy (Eq. 2–3). Fig. 3 further visualizes measurement-space difference maps, with extrinsic cases showing much lower errors than intrinsic ones within hallucinated regions.

Table 2. **Hallucination taxonomy compliance of HalluGen using Mean Squared Error within masked region.** Intrinsic sustains high measurement loss while extrinsic maintains low measurement loss despite semantic errors in image space. This validation confirms HalluGen faithfully generates both taxonomy types.

Predicted Region	Meas. Loss	Image Loss
DPS	0.006	0.022
Intrinsic	0.039	0.058
Extrinsic	0.003	0.041

4.3. Controllability of HalluGen

HalluGen enables fine-grained control over hallucination characteristics. We validate controllability along three dimensions: gradient strength, patch count, and patch size. We focus on intrinsic hallucinations, as extrinsic generation involves balancing multiple loss terms that complicate the isolation of individual control factors.

Severity control via gradient strength. Figure 4 (left) shows that hallucination severity increases with gradient ascent strength, γ . The squared error in hallucinated regions increases from 718 ($\gamma = 0.0005$) to 767 ($\gamma = 0.01$), showing that stronger gradients induce more severe, measurement-inconsistent hallucinations. FID remains stable, indicating preserved perceptual realism.

Spatial scalability via number of patches. Figure 4 (middle) validates that HalluGen scales with the number of patches. As the number increases from 1 to 5, total squared error grows approximately linearly (150 to 810). FID remains consistently low, confirming that distributed hallucinations maintain realistic appearances.

Patch size flexibility. Figure 4 (right) shows controllability across different spatial granularities. Squared error scales with patch area (16×16 : 488, 64×64 : 9000), while FID remains stable, demonstrating consistent realism across fine-grained and coarse spatial scales.

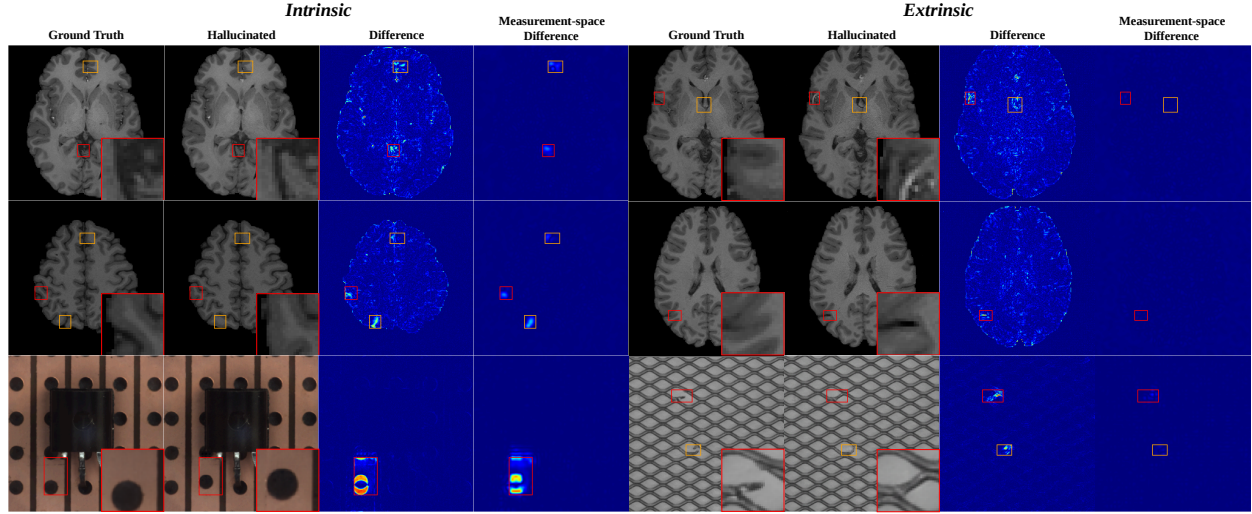


Figure 3. **Representative hallucinations generated by HalluGen across domains.** Top: Controlled synthesis of intrinsic and extrinsic hallucinations in our open dataset for MR images. Bottom: Cross-domain generalization to industrial imagery (MVTEC AD). HalluGen produces realistic yet semantically incorrect features across domains. More visual results are in the supplementary.

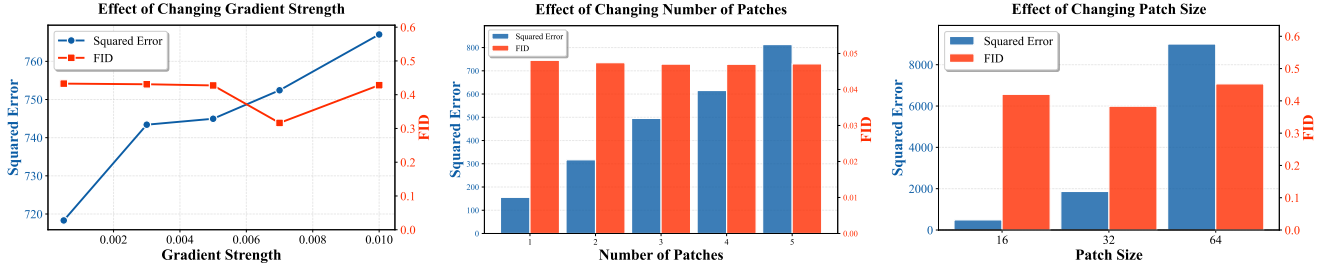


Figure 4. **Controllability of HalluGen.** Left: Severity increases with gradient strength γ . Middle: Severity scales linearly with number of patches while FID stays low. Right: Stable realism across patch sizes (16×16 – 64×64). HalluGen provides fine-grained control over severity, spatial extent, and granularity while preserving realism.

These results show that HalluGen enables precise control over hallucination severity, extent, and scale while preserving perceptual realism through manifold regularization.

4.4. Cross-Domain and Cross-Task Generalization

We evaluate HalluGen’s generalizability by sampling 200 images per domain-task pair (1600 total) for both intrinsic and extrinsic, summarized in Fig. 6. For MVTEC AD [5], semantic deviation is measured by the change in anomaly scores between ground-truth and generated images using the pre-trained PatchCore [38]. On ImageNet [12], it is quantified by the CLIP score, the distance between CLIP embeddings of the ground-truth and generated images.

Fig. 3 shows HalluGen generating hallucinated features on MVTEC AD, such as missing transistor legs and distorted grid patterns. It maintains low FID with large change in anomaly score (Fig. 6), indicating perceptually plausible but semantically divergent outputs. HalluGen also generalizes to natural images, generating realistic distortions such as extra insect legs, warped tires with an increase in CLIP score compared to DPS (Figs. 5, 6).

Additional experiments on HCP [42] confirm robustness across standard restoration tasks, including super-resolution and deblurring, showing that HalluGen generalizes across both domains and restoration types.

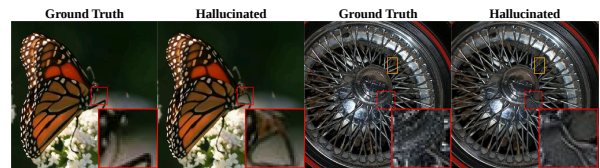


Figure 5. **Visual results of HalluGen on ImageNet.** HalluGen generalizes beyond safety-critical domains and can generate realistic hallucinated features as highlighted in red and orange boxes.

4.5. Case Study I: Hallucination Metric Benchmark

The evaluation of hallucination has been hindered by the lack of labeled data. Leveraging HalluGen-generated dataset, we establish the first benchmark for evaluating hallucination sensitivity of full-reference image quality metrics. We assess representative metrics by discriminabil-

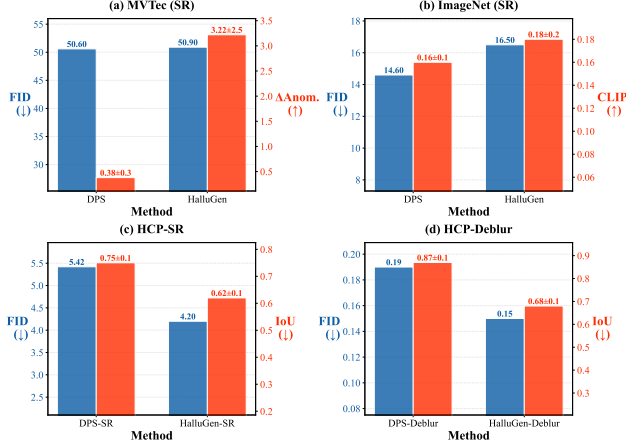


Figure 6. **Cross-domain and cross-task generalization of HalluGen.** Low FID and high semantic deviations confirm HalluGen’s generalizability. For SR, we use $\times 6$; for deblur, $\sigma = 3.0$.

ity (Cohen’s d), detectability (AUC-ROC), sharpness bias curve (SBC), and correlation with hallucination severity (Spearman’s ρ). SBC is computed as the area under the win-rate curve obtained by progressively low-pass filtering non-hallucinated images. Severity is defined as the sum of squared error within hallucinated regions scaled by the number of hallucinated patches, to penalize images with more frequent hallucination occurrences.

We also develop a hallucination-aware baseline, *Semantic Hallucination Assessment via Feature Evaluation* (SHAFE), to address the limitation of conventional metrics on hallucinations and evaluate it alongside existing metrics.

Semantic Hallucination Assessment via Feature Evaluation (SHAFE). Most full-reference metrics uniformly average errors, which dilutes hallucinations that are *localized and sparse*. SHAFE applies soft attention-based aggregation [50], emphasizing rare but structurally inconsistent regions while preserving global context. Given prediction and ground truth images, SHAFE is computed as:

- **Patch-level feature discrepancy.** A low-pass filter is applied to both inputs to suppress high-frequency noise and emphasize structural content. Each image is then fed into a pretrained vision encoder to extract semantic features, and cosine distance δ_{cos} is computed for each patch. Shallow multi-layer features are extracted to retain spatial richness and reduce dataset-specific bias of the encoder.

- **Softmax-based aggregation.** These patch scores are then aggregated using temperature-controlled softmax:

$$\text{SHAFE} = \sum_i w_i \delta_{cos,i}, \quad w_i = \frac{\exp(\delta_{cos,i}/\tau)}{\sum_j \exp(\delta_{cos,j}/\tau)}. \quad (11)$$

Smaller τ assigns exponentially greater weights, w_i , to local and severe errors. Its patch-based design enables hallucination localization, with Fig. 7 showing SHAFE heatmaps

accurately highlighting real failure regions.

SHAFE operates in feature space, but its formulation is general and can be applied to other metrics. Moreover, its differentiability allows integration into supervised training pipelines, opening avenues for future extensions toward hallucination-aware model optimization. See the supplementary material for more details.

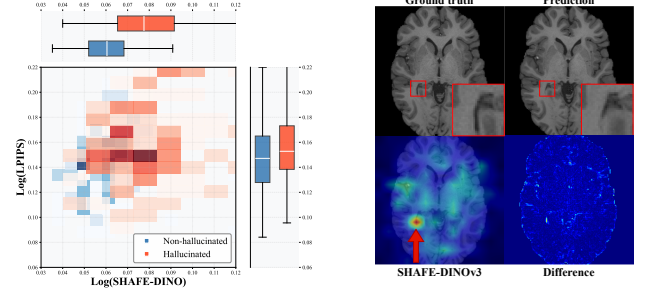


Figure 7. **Bivariate distributions of SHAFE-DINOv3 vs. LPIPS (left) and Spatial localization ability of SHAFE on real prediction (right).** Hallucinated examples shift toward higher SHAFE values, indicating improved separability compared to LPIPS.

Perceptual metrics are insensitive to hallucinations; Tab. 3 summarizes the performance of the metrics on hallucination. LPIPS and DISTS perform marginally above chance in hallucination detection, showing limited sensitivity. Their *low* Cohen’s $|d|$ and *high* per-sample variability reveal inconsistent responses across hallucinations, making them unreliable indicators. Because they uniformly average spatial errors and rely on ImageNet-trained features, these metrics favor perceptual sharpness over semantic correctness and often overlook structurally inconsistent regions.

Multi-scale improves sensitivity. MS-SSIM substantially improves hallucination sensitivity over SSIM, benefiting from its multi-scale formulation that captures hallucinations of varying sizes. This demonstrates the value of incorporating multi-scale information for more reliable image evaluation. However, it still underperforms SHAFE due to its uniform aggregation strategy.

Pixel metrics show type-dependent performance. Pixel-based metrics achieve moderate detection on intrinsic hallucinations that violate measurement consistency and produce large pixel-space deviations. However, their performance degrades substantially on extrinsic cases that remain measurement-consistent despite semantic incorrectness, limiting their utility for comprehensive evaluation.

SHAFE achieves superior performance. SHAFE variants outperform baselines, with *all feature backbones* improving all benchmark metrics by 30% over the best baselines. Especially, SHAFE-ResNet50 outperforms LPIPS and DISTS, despite sharing the same pre-training domain, highlighting the effectiveness of our weighted aggregation to emphasize local failures. Moreover, all SHAFE variants

Table 3. **Comprehensive hallucination benchmark across metrics.** Comparison of pixel- and feature-based metrics on effect size (discrimination), AUC (detection), Sharpness Bias Curve (sharpness bias), and severity correlation (spearman rank correlation). The rightmost column shows AUC on raw predictions with manually labeled (binary) hallucinations. Baseline metrics show weak hallucination sensitivity, whereas SHAFE improve detection and interpretability. FPR/FNR are calculated using the optimal threshold from AUC curve.

Metric	Type	Effect Size (Cohen’s $ d $) $\uparrow$		AUC-ROC $\uparrow$		SBC $\uparrow$		Severity (ρ) $\uparrow$		FNR/FPR (%) $\downarrow$	Real AUC $\uparrow$
		Intr.	Extr.	Intr.	Extr.	Intr.	Extr.	Intr.	Extr.		
PSNR	Pixel	0.33 $\pm$ 0.95	0.02 $\pm$ 0.99	0.59	0.51	0.76	0.49	0.37	0.32	62 / 28	0.58
SSIM	Pixel	0.09 $\pm$ 0.99	0.33 $\pm$ 1.03	0.52	0.40	0.22	0.08	0.11	-0.17	99 / 0	0.53
MS-SSIM [48]	Pixel	0.69 $\pm$ 1.00	0.16 $\pm$ 0.99	0.79	0.21	0.84	0.27	0.43	0.08	36 / 57	0.57
SCC [54]	Pixel	0.06 $\pm$ 1.03	0.16 $\pm$ 1.01	0.48	0.46	0.23	0.12	0.05	0.02	80 / 18	0.47
LPIPS [53]	Feature	0.03 $\pm$ 1.00	0.01 $\pm$ 0.99	0.51	0.50	0.14	0.12	0.11	0.14	26 / 70	0.42
DISTS [13]	Feature	0.06 $\pm$ 0.99	0.10 $\pm$ 0.99	0.52	0.54	0.27	0.27	0.19	0.12	31 / 60	0.44
VIF [40]	Pixel	0.29 $\pm$ 1.00	0.00 $\pm$ 0.99	0.58	0.50	0.28	0.11	0.27	0.17	42 / 48	0.47
SHAFE-ResNet50 [52]	Feature	1.18$\pm$1.12	0.84 $\pm$ 1.28	0.82	0.72	0.79	0.71	0.51	0.57	27 / 25	0.58
SHAFE-MedSAM [9]	Feature	1.13 $\pm$ 1.26	0.93$\pm$1.34	0.81	0.75	0.77	0.71	0.47	0.60	32 / 14	0.70
SHAFE-DINOv3 [41]	Feature	1.12 $\pm$ 1.17	0.92 $\pm$ 1.30	0.76	0.74	0.71	0.67	0.51	0.58	30 / 26	0.79

exhibit low sharpness bias, confirming that the combination of low-pass filtering and our multi-layer feature extraction strategy reduces the reliance on high-frequency artifacts.

We evaluated metrics on real restoration outputs by annotating 300 predictions from SwinIR [31], EDSR [32], and ESRGAN [45]. SHAFE shows much stronger discrimination, improving AUC by 0.2 over traditional metrics (Table 3), aligning more closely with human judgment. Figure 7 shows that SHAFE provides distinctly better separation between images with and without hallucinations than LPIPS, demonstrating its robustness in real cases.

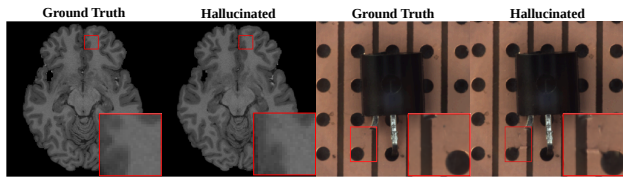


Figure 8. **Failure case of HalluGen in homogeneous regions.** In smooth regions (e.g., brain putamen or background), diffusion prior dominates, leading to visually weak hallucinations.

4.6. Case Study II: Hallucination Detection

HalluGen’s utility can be extended to developing data-driven hallucination detector. We train a *reference-free hallucination detector* using a CNN that takes the prediction $\hat{x}$ and measurement y as input and predicts whether hallucinations are present, without requiring ground-truth images. The model is trained on the full HalluGen dataset (4,350 images) and evaluated on: (1) a held-out HalluGen split (702 images; evenly balanced across intrinsic, extrinsic, and non-hallucinated cases), and (2) same real restoration outputs used in the benchmark in Tab. 3.

As shown in Table 4, the detector achieves strong performance on synthetic data and generalizes to real predictions, despite its simple architecture. Detection is notably easier for intrinsic hallucinations than extrinsic by AUC $\approx$ 0.14.

This is because intrinsic hallucinations break measurement consistency, whereas extrinsic ones remain consistent with y and are therefore harder to identify. Overall, these results show that HalluGen *transfers to real model failures* and enables comprehensive, systematic hallucination evaluation that was previously infeasible.

Table 4. **Reference-free hallucination detection.** CNN trained only on HalluGen; thresholds selected on validation.

Test Set	AUC $\uparrow$	F1 $\uparrow$
HalluGen (Intrinsic)	0.91	0.95
HalluGen (Extrinsic)	0.77	0.76
Real Predictions	0.73	0.63

5. Conclusion

We present *HalluGen*, a principled framework for controllable hallucination synthesis, and construct the first hallucination dataset for image restoration with patch-level annotations. We demonstrate HalluGen’s utility in two key applications: (1) establishing a hallucination-metric benchmark that reveals the failure of existing image quality metrics and developing *SHAFE*, which improves hallucination sensitivity via soft-attention pooling over feature errors; and (2) training reference-free hallucination detectors that generalize to real restoration outputs. These results demonstrate that HalluGen *transfers to real model failures, enabling systematic study of hallucinations in real-world settings*. HalluGen has several limitations, including difficulty inducing hallucinations in homogeneous regions due to strong diffusion priors (Fig. 8) and lack of explicit semantic control. The current dataset only features healthy MR images, and extending HalluGen to diverse pathologies remains a priority. Nevertheless, HalluGen establishes the first systematic foundation for generating, detecting, and evaluating hallucinations in image restoration. We release our code, dataset, and metrics to support further research and safer deployment in safety-critical domains.

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